

when do you feel the
most like a CS student?

CS 340

Debugging + Terminal

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Q1

~Code~
340



Updates

1. MP 1.1

Do them soon
→ Sept 15

2. Environment check-off

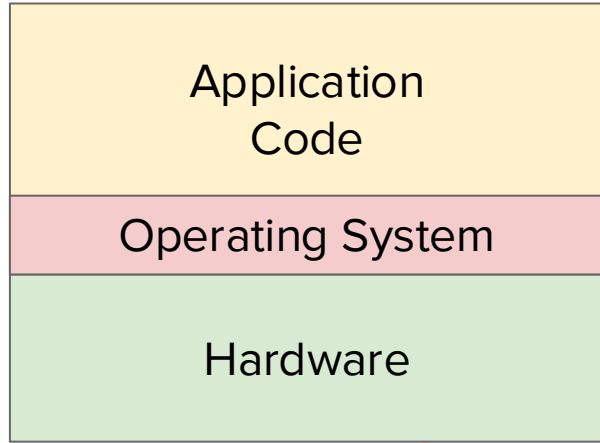
3. HW 1 — out today
due next week wed.

4. Collaboration time/office hours today! 3:30 - 4:30

Agenda

1. Review
2. Terminal/Shell Overview
 - a. Basic commands
3. VScode Debugger
4. Docker
5. VM

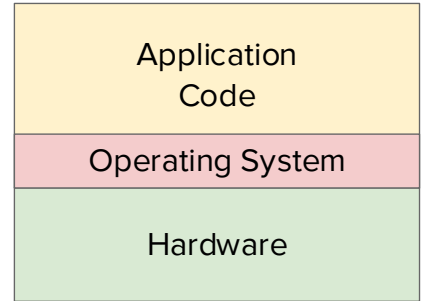
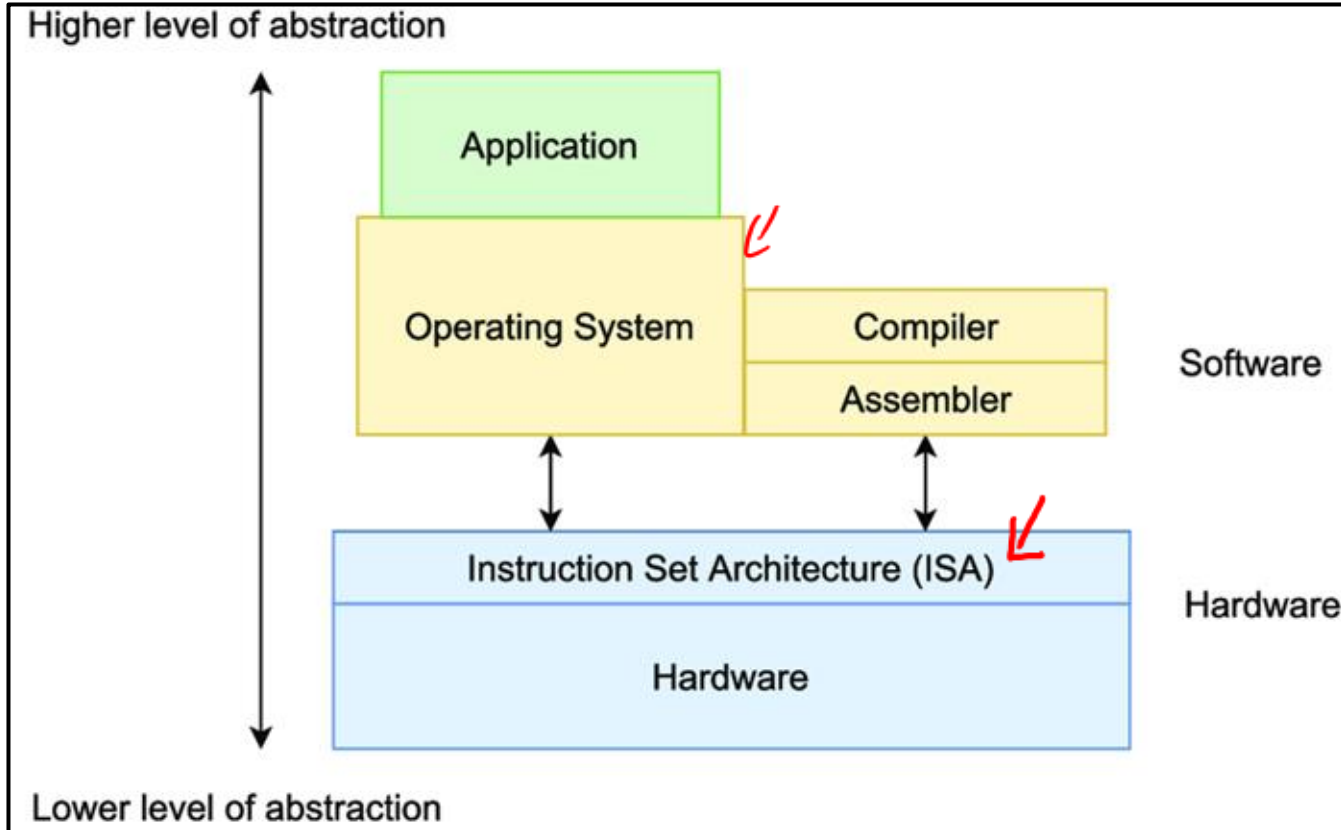
Big Ideas



VScode, compiler, chrome, shell
manages what code is running, moves bytes around, permissions
interprets machine instructions



What is misleading about this image?



Terminal/Shell + Debugging + Docker/VM

LG: Be able to compile, run, and debug C++ and C programs.

LG: Have a big picture understanding of your local file system, the docker environment, and your assigned virtual machine.

Mental models we are developing:

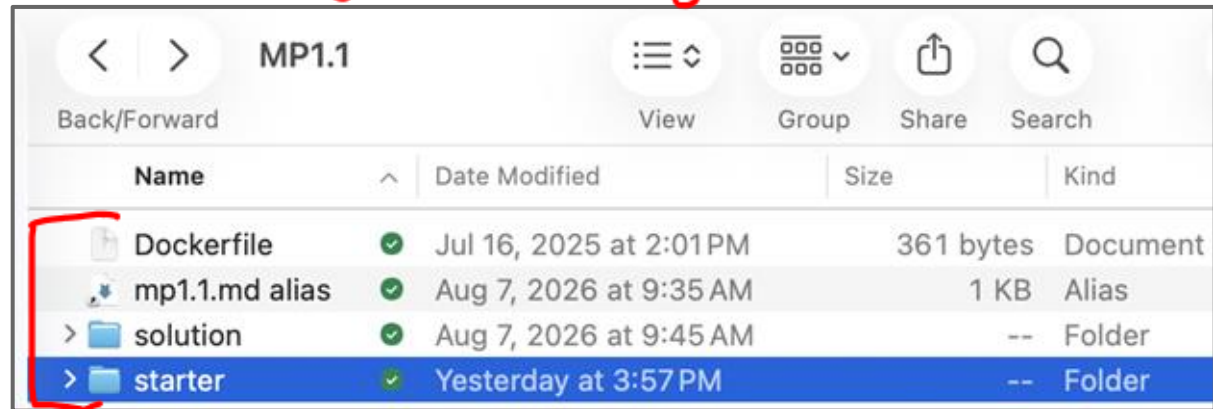
How a file system works on a computer

Compile/execute flow for compiled languages

What a debugger workflow looks like

Interacting with a computer

#1 click things



Name		Date Modified	Size	Kind
Dockerfile	✓	Jul 16, 2025 at 2:01PM	361 bytes	Document
mp1.1.md alias	✓	Aug 7, 2026 at 9:35 AM	1 KB	Alias
> solution	✓	Aug 7, 2026 at 9:45 AM	--	Folder
> starter	✓	Yesterday at 3:57 PM	--	Folder

#2 terminal running a shell program

```
drschatz@cs-drschatz-C35F43D73N MP1.1 % ls
Dockerfile      solution
mp1.1.md alias  starter
drschatz@cs-drschatz-C35F43D73N MP1.1 %
```

Basic Commands

`pwd` - prints your current directory path

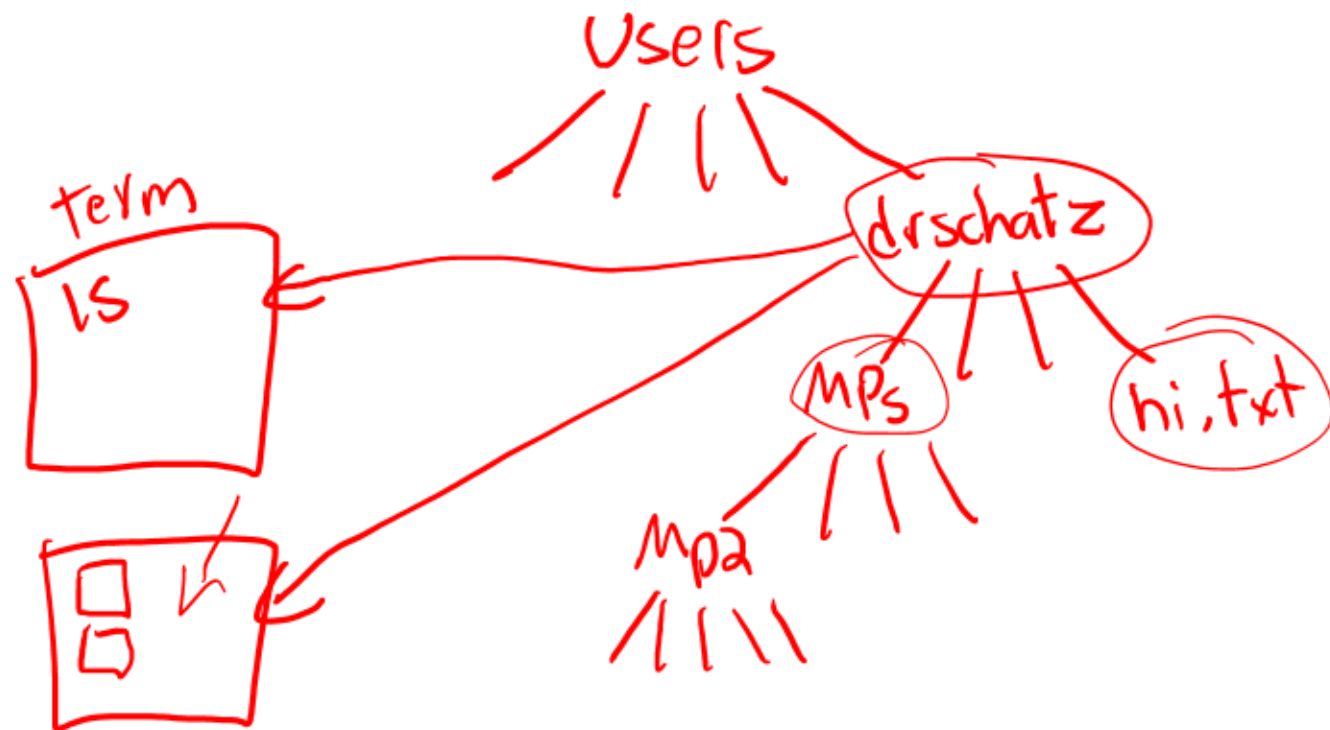
`ls` - prints the files and directories in your current directory

`cd` - moves you to a different directory

folder == directory

path - series of directories, separated by /

File Structure Example



What would pwd print on the next line?

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```
drschatz@cs-drschatz-MBP 340-Sp26 % pwd  
/Users/drschatz/Documents/340-Sp26  
drschatz@cs-drschatz-MBP 340-Sp26 % cd ../../
```

Pwd

Running a python program

python3 test.py
 ↑
 path

What goes in the box to run the python code?

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```
drschatz@cs-drschatz-MBP term-demo % ls
Makefile      test_c.c      test_python.py
test_c        test_c.h      use_test_c.c
drschatz@cs-drschatz-MBP term-demo % pwd
/Users/drschatz/Documents/340-Sp26/cs340/term-demo
drschatz@cs-drschatz-MBP term-demo % cd ../
drschatz@cs-drschatz-MBP cs340 % ls
Dockerfile    mp1           term-demo
drschatz@cs-drschatz-MBP cs340 % cd mp1
drschatz@cs-drschatz-MBP mp1 % python3 test_python.py
```

../term-demo/

../cs340

cs340/mp1

Big Idea

You can use a terminal running a shell to -

Navigate your computer's files
run applications
python
compiler
ssh, scp

Running a C/C++ program

1. Compile

`cc main.c func.c -o exec`

1. Run executable file

`./exec`

Compiling Details

1. Compile

a. Each .c file listed

- copy includes
- check every func has been declared
- .o files

b. Link together .o files

- checks for 1 definition for each func

I edit the spelling of a print statement in my C program, I need to recompile the file to see the change.

- ☒ A) True
- ☐ B) False
- ☐ C) Unsure

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MP 1.1

`char *ptr = 0;`

↑
type

`*ptr = 4;`

follow
address
and
assign to 4

ptr
0

0x6780

4

Challenge: How many would cause segfaults?

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a) char c = 0; char *ptr = &c; *ptr = 4;

b) char *ptr = 0; *ptr = 4;

c) char *ptr = 0; ptr = 4;

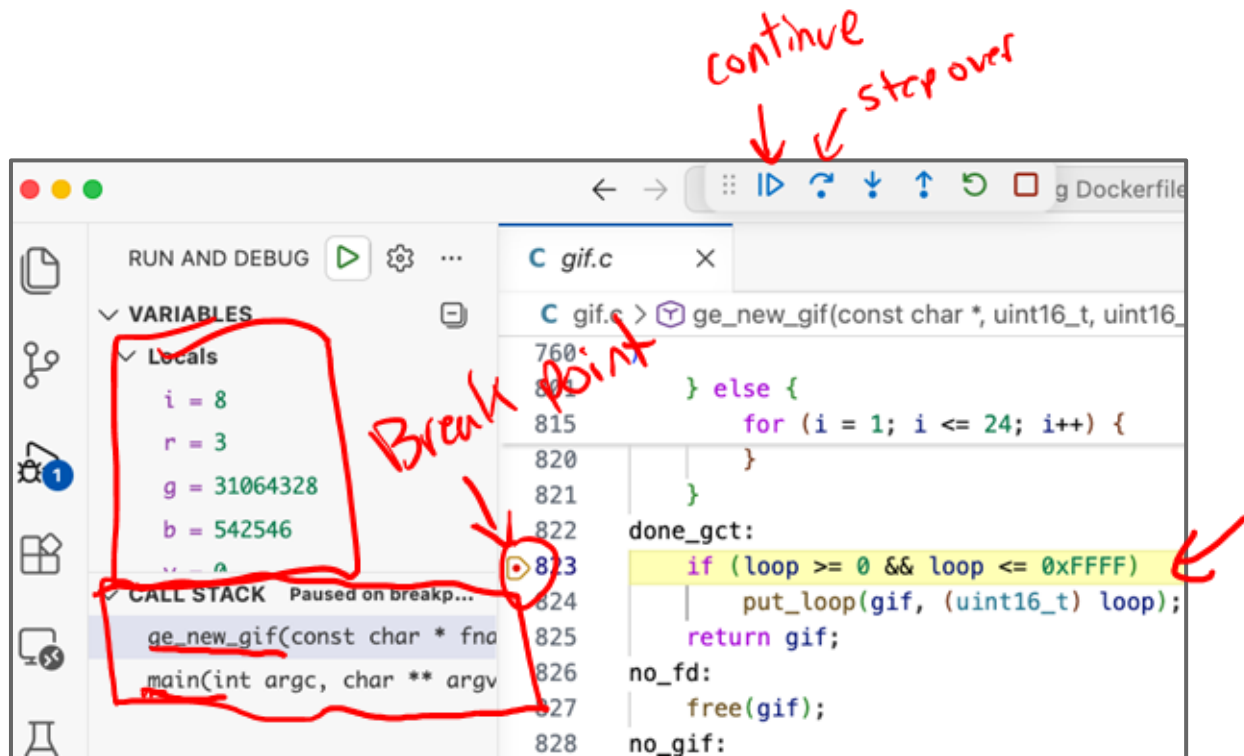
d) char *ptr = 0; ptr = 4; *ptr = 0;

seg fault



2

VS Code Debugger Overview



Docker

Creates an *a light weight linux environment*
on your local computer. Mounts to your local file system so all
changes *persist across both*

-uses your computer hardware

I make a file test.c while in my docker environment on VS code.

True or false, I can access test.c through my computer's local file system interface.

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True or False: Docker and my local computer may have different version numbers for the same application.

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Q7

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Docker Vs. VM

Docker - creates a lightweight linux environment on your local computer. Mounts to your local file system so all changes persist across both.

Virtual Machine - a slice of a computer on campus

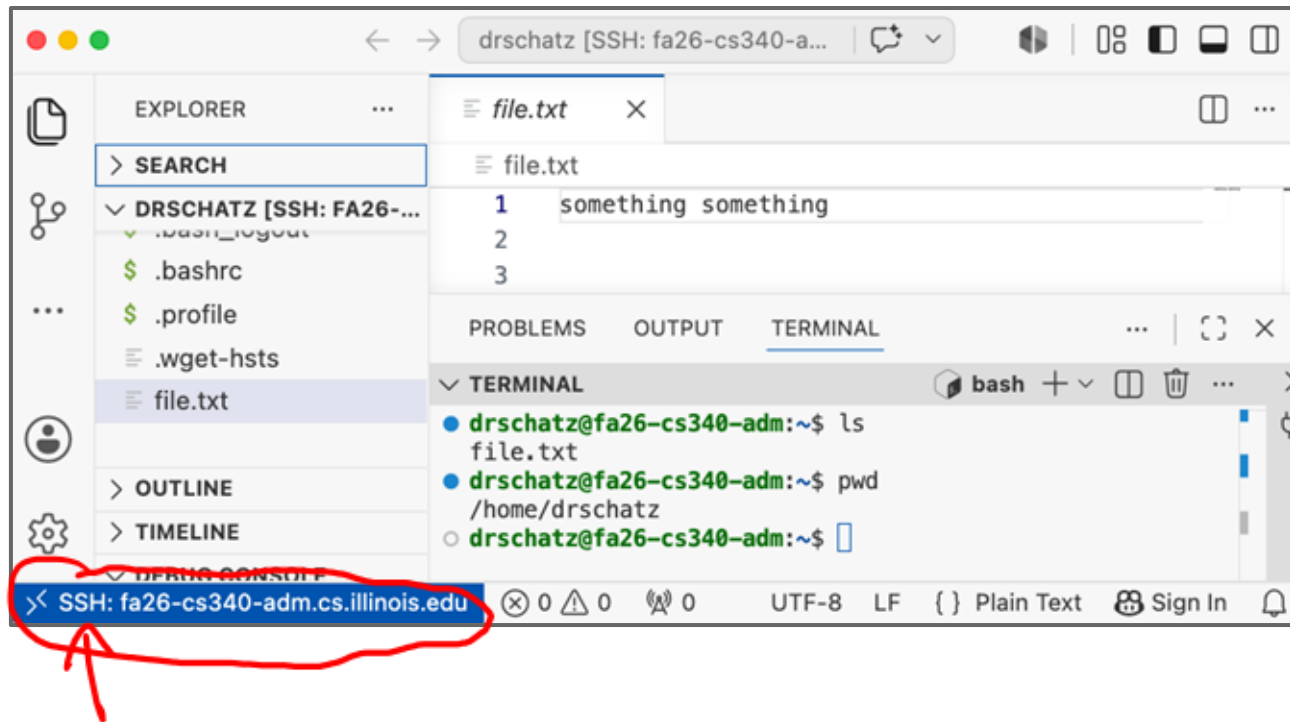


SSH

Creates a **connection** to another computer's shell process. You can use it to access your **VM**

% ssh drschatz@fa26-cs340-001.cs.illinois.edu
%
↑
exit()

SSH



SCP

SCP - **copies** the files on one computer to another

↓ ↓
scp mp1.1/solution.txt drschatz@fa26-cs340-adm.cs.illinois.edu:home/drschatz/cs340/

↑
current
location

↑
dest

↓

**True or false, I need to use SCP
to transfer files between my
docker environment and my
local files.**

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1. I make a file test.c while in my docker environment on VS code
2. I SCP test.c to my VM
3. I edit test.c while in my docker environment

True or false, test.c is now updated in both the VM and docker environment?

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- 1. I make a file test.c while in my docker environment on VS code**
- 2. I SCP test.c to my VM**
- 3. I edit test.c while in my docker environment**

True or false, test.c is now updated in both the docker environment and my local environment?

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Terminal/Shell + Debugging + Docker/VM

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Relevant C++ Review

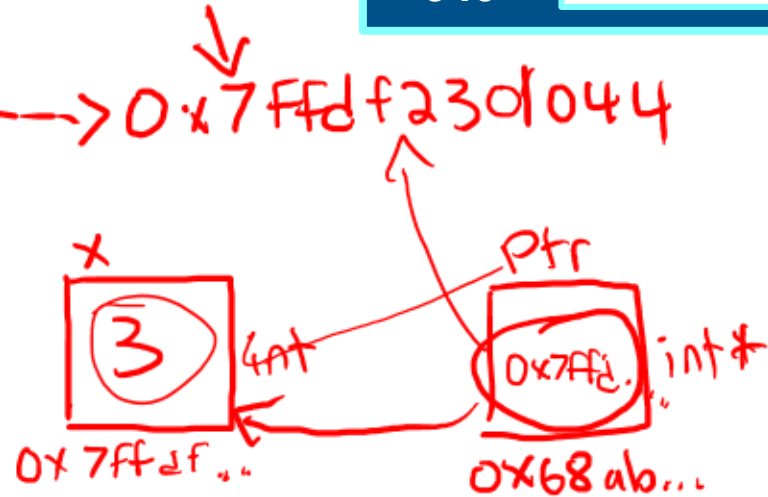
**What could print on lines 8 and 9?
(separated by a space)**

```
4  int main(){
5      int x = 3;
6      int *ptr = &x;
7      cout << ptr << endl;
8      cout << &ptr << endl;
9      cout << *ptr << endl;
10 }
```

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`0x68ab... 3`

What could print on lines 7 and 8?

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Q12

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```
4  int main(){  
5      int arr[3] = {1, 2, 3};  
6      cout << arr << endl;  
7      cout << &arr << endl;  
8      cout << *arr << endl;  
9  }
```

0x7ffdf23...

0x7ff... 1

$arr[2] == *(arr + 2)$



What could print on lines 7 and 8?

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```
4  int main(){  
5      int *arr = new int[4];  
6      cout << arr << endl;  
7      cout << &arr << endl;  
8      cout << *arr << endl;  
9      delete[] arr;  
10 }
```

→ 0x7ffd...
← 68ab
Stack
→ 0

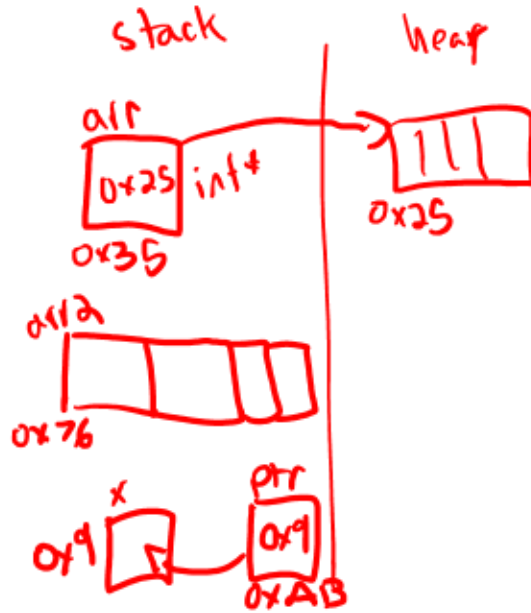
arr
int*
b
0x68ab
a
↑
0x68ab

heap



Summary

```
4  int main(){
5      int *arr = new int[4];
6      → int arr2[4];
7      → int x;
8      int* ptr = &x;
9      delete[] arr;
10 }
```



Data Stored as Bytes

Physically, computer hardware...

To make binary more practical we...

Data types have a size (in bytes)...

Addresses in a C/C++ Program

Pointer math